Make no mistake, the overall performance has improved by a wide margin over the years, but this technology still pales in comparison to other panel types. Mind you, a decent TN panel can still provide sharp and detailed picture with adequate contrast around the 1000:1 mark. However, the colour reproduction isn't as accurate as the competition. In fact, the overall vibrancy and colour performance is noticeably lacking. That, sadly, isn't even its main problem. Restricted viewing angles are by far its biggest failing. Don't go by the specs, where you can find generous viewing angles mentioned, but the amount of colour shift in practical terms is disconcertingly high. View these panels from reasonably off centre position and the colour shift gets worse to the point of complete inversion, wherein the image appears as a film negative. While controlling horizontal viewing angles is relatively easy in a monitor for a single user, controlling the vertical viewing angle is a lot more difficult. If you have a large enough monitor (say 27-inch), colour shifts are going to be a serious problem unless we're dealing with a really good TN panel.

In short, choose TN panels only if you value 3D and gaming over colour accuracy. These are definitely not the kind of panels you choose for colour accurate web designing, video production, and photo manipulation purposes.

IPS Panels

If TN is at one end of the LCD quality spectrum, IPS (In Plane Switching) panels lie at the very extreme. Nothing beats this technology when it comes to absolute picture quality. It delivers the trifecta of image quality, colour accuracy, and viewing angles, thereby making it indispensible for colour accurate work such as web/graphics design and photo/video editing. As expected, these monitors also happen to be at the higher end of the price spectrum. But for the money, you get sharp, crisp picture with vibrant colour reproduction and viewing angles as wide as 178 degrees. In fact, IPS panels are the way to go if you're looking for a larger display meant to accommodate more than one viewer.

Pricey IPS panels may offer much higher image fidelity than TN panels, but the technology still has its Achilles heel. The response time of this technology is the slowest of the lot, and ranges from 6 ms to 16 ms depending upon how deep your pockets can go. Make no mistake, this is only a tad bit slower than the average TN panel and not even noticeable in day to day usage. However, slower response time can be telling in videogames. If you're an FPS gamer who's particular about image